Lead Exposure in Childhood: A Review of Risks, Consequences, and Interventions

Prashanth Mada[1] and Radhika Soanker[2]
[1]Department of Forensic Medicine and Toxicology, [2]Department of Pharmacology,
All India Institute of Medical Sciences, Bibinagar, Telangana, India

INTRODUCTION

Lead is an extremely toxic element that can harm nearly every organ in a child's body. Children are significantly more vulnerable to the harmful effects of lead exposure compared to adults for several reasons. Firstly, infants and young children absorb four to five times more lead into their bodies than adults when exposed. Secondly, their play behaviour increases the risk of ingesting lead-contaminated soil and dust, as they frequently play on the ground and often put their hands or objects in their mouths. Finally, the blood-brain barrier in children is still in the process of developing, making them more susceptible to neurological damage from lead exposure, even at relatively lower levels. [1,2]

Blood lead levels in the general population have significantly declined following the global shift from leaded to unleaded gasoline. However, according to the World Health Organization, lead exposure can be dangerous at any level. What we once believed to be safe amounts are now known to harm children's health and brain development [2] Therefore this article presents the current understanding of the impact of lead exposure in children and the strategies to minimize the exposure.

Burden in India

A systematic review analyzing blood lead levels (BLLs) in low- and middle-income countries (LMICs) included studies published between 2010 and 2019. Of the 520 studies assessed, 41 originated from India. These studies encom passed pooled data from 5,593 children and 2,22,353 adults. The mean BLL was reported as 5.22 µg/dL in children and 4.42 µg/dL in adults. Notably, a significant disparity in childhood lead exposure was observed between LMICs and high-income countries. For instance, data from the 2017 CDC National Childhood Blood Lead Surveillance in the United States indicated that fewer than 2% of children aged 0–5 years had BLLs exceeding 5 µg/dL. [3]

A 2018 review article on the prevalence of elevated blood lead levels (BLLs) in children in India cited data from the 1998–1999 National Family Health Survey (NFHS), which was the first to report BLLs in children under 3 years of age in two major Indian metropolitan cities Mumbai and Delhi. According to the survey, approximately 50% of children in these cities had BLLs ≥10 µg/dL (NFHS, 2000). [4]

Sources of Lead:

Lead exposure in children can originate from a wide range of sources. Common sources include traditional cosmetics, lead-based paints and pigments, and some toys and jewellery. Certain herbal, traditional, and Ayurvedic medicines may also contain lead. Lead dust and chips from deteriorating paint are a frequent cause in older buildings. Some imported spices, candies, and food items may be contaminated. Additional sources include solders in food cans, lead-based ceramic glazes on dishes and cookware, and some metallic pots and pans. Leaded gasoline and old water pipes or fixtures are also risks, particularly in older communities. Industrial pollution, poorly regulated used lead-acid battery (ULAB) recycling sites, and emissions from waste incinerators contribute to environmental contamination. Soil where children play or food is grown may be tainted with lead, especially in high-risk areas. Lastly, family members who work in lead-related occupations may inadvertently bring home lead dust on their clothes and shoes, exposing others in the household. [1,2]

The systematic review conducted in low- and middle-income countries (LMICs) identified key sources of lead exposure, which included informal recycling and manufacturing of lead-acid batteries, metal mining and processing activities, electronic waste handling, and the intentional adulteration of food—particularly spices—with lead. [3]

Molecular Mechanism of Lead Effects

Lead interferes with how the nervous system works by disrupting calcium signals, which are vital for brain function. It can block or mimic calcium, allowing it to attach to important proteins and enzymes, changing how they work. This inter ference can stop or over-activate processes like the release of brain chemicals and activation of key proteins. All of these effects damage the development of the brain and nervous system, putting a child's healthy growth at serious risk. [1]

Clinical presentation of childhood lead poisoning

Childhood lead poisoning is difficult to detect because it often develops quietly. When lead exposure is low to moderate, children typically exhibit no symptoms, and signs are rarely apparent during clinical examination. With greater exposure, children may show vague symptoms such as headaches, fatigue, and stomach pain, while neurological signs like clumsiness or drowsiness may appear. Severe cases can lead to vomiting, stupor, and seizures. [1,2,5]

However, prolonged exposure may cause anaemia and affect the nervous system, leading to symptoms such as headaches, irritability, tiredness, seizures, muscle weakness, loss of coordination, shaking, and even paralysis. [1,2]

In addition, lead exposure before or during pregnancy can endanger both mother and fetus. Stored lead may be released from the mother's bones, especially with low calcium levels, increasing blood lead levels. This can lead to miscarriage, stillbirth, premature birth, low birth weight, and developmental problems affecting hearing, vision, and learning [1,2]

Effect on Neurological development

One of the most serious effects of lead in children is the harm it does to the developing brain. Even small amounts of lead can cause learning problems, lower IQ scores, trouble in school, and behavior issues. Children with blood lead levels as low as 5 µg/dL can score 3 to 5 points lower on IQ tests than children without lead exposure.(1,2) If this drop in IQ were to happen across an entire population, it could lead to a 57% rise in the number of people with intellectual disabilities. [1,5]

A comprehensive meta-analysis assessing childhood blood lead levels in India concluded that, on average, children may lose approximately 4 IQ points due to lead exposure. The analysis pooled data from 31 studies involving 5,472 participants from nine states. The mean blood lead concen tration was reported as 6.86 µg/dL in children and 7.52 µg/dL in adults with no occupational exposure. [1]

Exposure to lead can negatively affect a child's brain development, making it harder for them to manage emotions, control impulses, and avoid aggressive behaviour. These effects are often lasting and have been linked to a greater risk of violence and criminal behaviour later in life. [1]

Research in the United States has shown that higher blood lead levels in childhood can predict a greater likelihood of being arrested for violent crimes as an adult. A 2017 study involving 12,000 children born between 1990 and 2004 found that for every 1 µg/dL increase in lead levels, the chance of school suspension rose by 6.5% to 7% for boys and 6.4% to 9.3% for girls. [1]

Treatment of Health Effects

Unfortunately, there is currently no definitive cure for lead poisoning. Once lead remains in the body for an extended period, it tends to accumulate in the bones, making it extremely difficult to eliminate. By the time this occurs, significant neurodevelop mental damage may already be irreversible. In cases of acute or severe lead poisoning, treatment may involve **chelation therapy**. This process uses chelating agents that bind to heavy metals—such as lead, mercury, arsenic, and iron—in the blood stream, allowing the body to excrete them through urine. [1] Succimer (dimercaptosuccinic acid) is a

water-soluble, oral chelating agent and calcium disodiumedetate (CaNa2 EDTA) is given by continuous intravenous infusion

However, chelation therapy is generally **not recommended** when blood lead levels are **below 45 µg/dL**, due to the risk of potentially serious side effects. [1,4]

Prevention of lead exposure

Protecting children from lead exposure begins with identifying potential risks in their surroundings. Older homes, schools, and buildings may contain lead-based paint, particularly if the paint is deteriorating or peeling. Additionally, water pipes and plumbing fixtures can be hidden sources of lead contamination. Children should be kept away from areas suspected of lead exposure. Encouraging regular hand and face washing—especially after outdoor play—helps minimize the risk of ingesting lead particles. A well-balanced diet rich in iron, calcium, and vitamin C can reduce the body's absorption of lead in case of exposure. Parents and caregivers who work in environments involving lead should take extra precautions by changing clothes and shoes before entering the home to prevent bringing lead dust inside. If there is any suspicion of exposure, it is essential to seek medical evaluation and specifically request a blood lead level test for the child. [1,2,4]

Strategies to minimise lead exposure

Blood lead levels in the general population have significantly declined following the global shift from leaded to unleaded gasoline. Additional strategies to reduce lead exposure include the removal of lead-based paints from households, strict regulation and oversight of recycling and smelting industries, and the rehabilitation of lead-contaminated environments. Building national capacity for blood lead level monitoring is essential to identify high-risk zones. Strengthening healthcare systems is also critical, enabling early detection, monitoring, and treatment of lead exposure—particularly in children. Furthermore, continuous public education campaigns should be implemented to raise awareness about the risks and sources of lead, targeting parents, schools, community leaders, and healthcare professionals alike. [1,2]

Legislation and policy

To effectively reduce lead exposure and its health risks, it is crucial to establish and enforce comprehensive environmental, health, and safety standards specifically targeting lead battery manufacturing and recycling operations. Legislation should also be developed and strengthened to regulate e-waste recycling, a growing source of lead contamination. Policies must be reinforced to eliminate the use of lead compounds in paints. Additionally, strict enforcement of environmental and air-quality regulations is essential for smelting industries. National drinking water standards must include specific parameters for lead to ensure safe and clean water for all communities.

Conclusion

Children are especially vulnerable to lead exposure, who may show no symptoms at low levels. Yet, such exposure is linked to reduced IQ, academic difficulties, and behavioural problems. Protecting children requires strict enforcement of environmental and public health regulations to eliminate exposure to potential sources of lead.

References

1. UNICEF and Pure Earth. The Toxic Truth: Children's Exposure to Lead Pollution Undermines a Generation ofFuture Potential. 2020. https://www. unicef.org/reports/toxic-truth-childrens-exposure-to-lead-pollution-2020 (Accessed May 30, 2025)
2. World Health Organization. Exposure to lead: a major public health concern. Preventing disease through healthy environments. 2019.https://www.who.int/publications/i/item /9789240037632 (Accessed June 2, 2025)
3. EricsonB, Nash E, Ferraro G, Sinitsky J, Taylor MP.Blood lead levels in low-income and middle-income countries: a systematic review. Lancet Planetary Health 2021;5 (3):e145-153
4. Parween A, Khan MM, Upadhyay T, Tripathi RV. Prevalence of Elevated Blood Lead Level in Children of India. Nature Environment and Pollution Technology. 2018;17(3):703-10
5. Councilon Environmental Health. Prevention of Childhood Lead Toxicity. Pediatrics. 2016;138(1):e20161493